

Piecewise Linear Approach for Solving Nonlinear Dynamic Systems

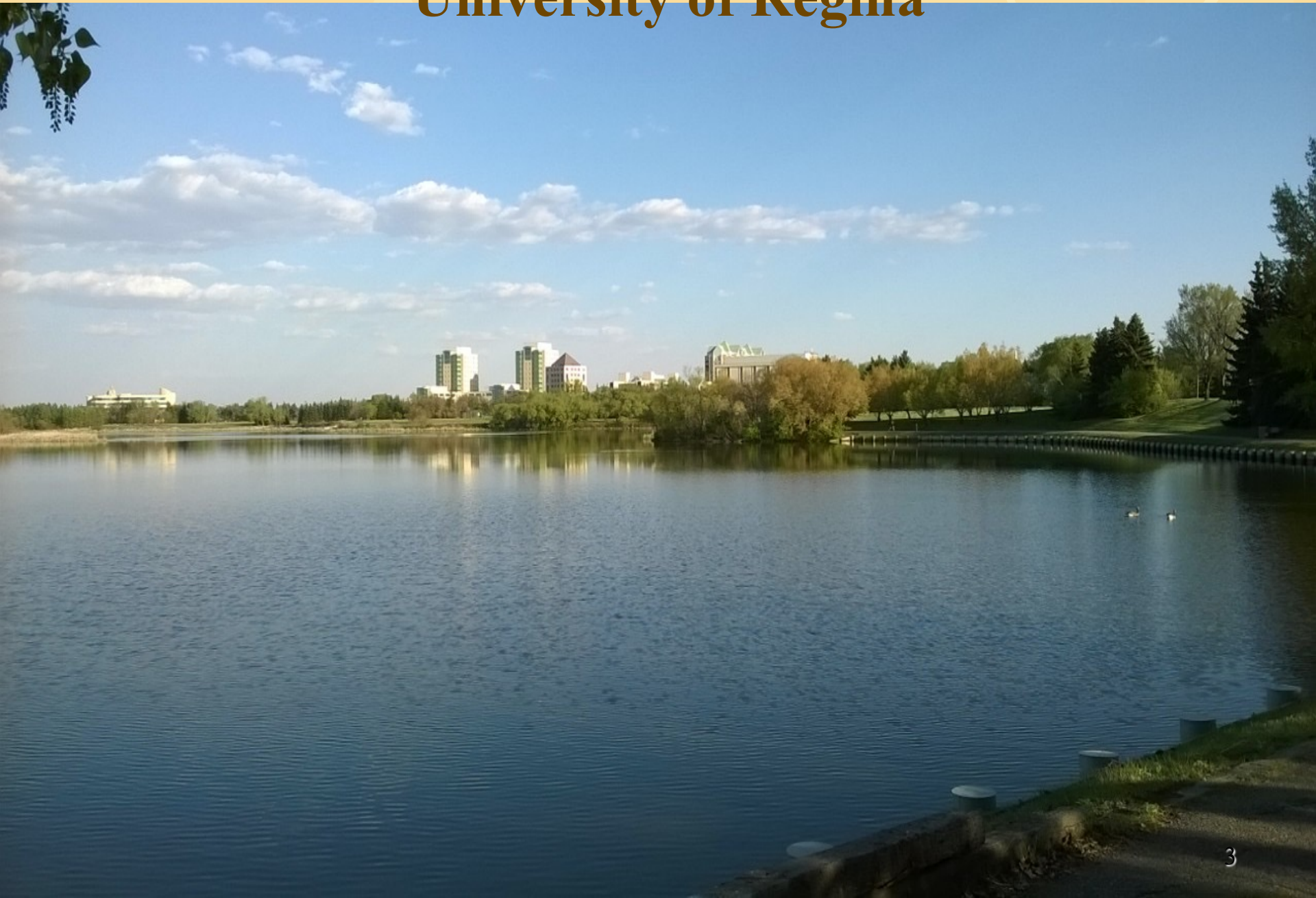
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Linear and Nonlinear Systems

- Nonlinear systems in various engineering areas
- Linearization and simplifications and nonlinear systems
 - Constitutive relations
 - Friction
 - Thermal extension
 - Governing equations of dynamics

Solving nonlinear dynamics problems

- Analytical solution attempts
- Semi-analytical solutions
- Approximate solutions
- Numerical solutions

Physical and mathematical modeling of piecewise linear systems

$$\frac{dy}{dx} = f([x])$$

$$\frac{d^n y}{dx^n} = g([x])$$

Discrete and piecewise constant

Governing equations of dynamic systems

Continuous systems

$$\ddot{x} + 2c\dot{x} + \omega^2 x = \varphi(t) + \psi(t, x, \dot{x})$$

PC/PL dynamic systems

$$m\ddot{x} + c\dot{x} + kx = f(x([t]))$$

Solution $x(t)$ satisfies the following conditions

- the displacement $x(t)$ and its derivative are continuous over the time domain;
- the derivative of the displacement exists at each point, with the possible exception at the integer points at which left-hand side derivatives exist;
- the solution satisfies equation on each time $n \leq t < n + 1$ interval, where n is an integer point in the domain.

Solution of damped spring-mass system

$$m\ddot{x} + c\dot{x} + kx = Ax([t])$$

In an arbitrary interval $n \leq t < n+1$ the general solution

$$x_n(t) = e^{-\theta(t-[t])} \left\{ \begin{aligned} & d_n \left(1 - \frac{A}{k}\right) \cos(\xi t - \xi[t]) \\ & + \frac{1}{\xi} \left[v_n + \theta d_n \left(1 - \frac{A}{k}\right) \right] \sin(\xi t - \xi[t]) \end{aligned} \right\} + \frac{A}{k} d_n$$

in which

$$\theta = \frac{c}{2m}$$

$$\xi = \left(\frac{k}{m} - \frac{c^2}{4m^2} \right)^{\frac{1}{2}}$$

$$\frac{k}{m} > \frac{c^2}{4m^2}$$

Conditions of continuity

$$x_n(n) = x_{n-1}(n)$$

and

$$\dot{x}_n(n) = \dot{x}_{n-1}(n)$$

Solution in an interval

$$\begin{bmatrix} d_n \\ v_n \end{bmatrix} = e^{-\theta t} \times$$

$$\begin{bmatrix} (1 - A/k)[\cos \xi + (\theta \sin \xi)/\xi] + (Ae^\theta)/k & \sin \xi / \xi \\ -(1 - A/k)[(\theta^2 / \xi + \xi) \sin \xi & \cos \xi - (\theta \sin \xi)/\xi \end{bmatrix} \begin{bmatrix} d_{n-1} \\ v_{n-1} \end{bmatrix}$$

$$n \leq t < n + 1$$

Complete solution

$$x(t) = \left\{ \begin{array}{l} (1 - \frac{A}{k})[\cos(\xi t - \xi[t])] \\ + \frac{\theta}{\xi} \sin(\xi t - \xi[t]) \end{array} \right\} + \frac{A}{k} e^{\theta(t-[t])} \frac{1}{\xi} \sin(\xi t - \xi[t]) \Bigg\} D_1$$

$$D_1 = e^{-\theta t} \left[\begin{array}{l} (1 - A/k)[\cos \xi + (\theta \sin \xi) / \xi] \\ + (Ae^\theta) / k \\ -(1 - A/k)[(\theta^2 / \xi + \xi) \sin \xi] \end{array} \right] \left[\begin{array}{l} \sin \omega / \omega \\ \cos \xi - (\theta \sin \xi) / \xi \end{array} \right]^{[t]} \left[\begin{array}{l} d_0 \\ v_0 \end{array} \right]$$

Solving linear dynamics problems

Introduction

$$[Nt] / N$$

Proof

$$\lim_{N \rightarrow \infty} \frac{[Nt]}{N} = t$$

Existence and uniqueness of solution

Solving linear dynamics problems

Implementation of PC argument: Analytical Solutions of linear dynamic systems

$$\ddot{x}(t) + \omega^2 x(t) = 0$$

Corresponding PL system

$$\ddot{X}_i(t) + \omega^2 X_i\left(\frac{i}{N}\right) = 0$$

$$i / N \leq t < (i + 1) / N, \quad i = 0, 1, 2, 3, \dots, [Nt]$$

Solving linear dynamics problems

Solution of PL system

$$X(t) = \begin{bmatrix} 1 - \frac{\omega^2}{2} \left(t - \frac{[Nt]}{N} \right)^2 & t - \frac{[Nt]}{N} \end{bmatrix} \begin{bmatrix} 1 - \omega^2 / 2N^2 & 1/N \\ -\omega^2 / N^2 & 1 \end{bmatrix}^{[Nt]} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix}$$

Let

$$\mathbf{Q} = \begin{bmatrix} 1 - \frac{\omega^2}{2N^2} & \frac{1}{N} \\ -\frac{\omega^2}{N} & 1 \end{bmatrix}$$

Solving linear dynamics problems

$$\mathbf{Q} = \mathbf{E}\mathbf{D}\mathbf{E}^{-1}$$

where $\mathbf{E}$ is a set of linearly independent eigenvectors of $\mathbf{Q}$ and $\mathbf{D}$ a diagonal matrix of the eigenvalues of $\mathbf{Q}$

$$\mathbf{Q}^{[Nt]} = \mathbf{E}\mathbf{D}^{[Nt]}\mathbf{E}^{-1} = \mathbf{E} \begin{bmatrix} \phi_1 & 0 \\ 0 & \phi_2 \end{bmatrix}^{[Nt]} \mathbf{E}^{-1} = \mathbf{E} \begin{bmatrix} \phi_1^{[Nt]} & 0 \\ 0 & \phi_2^{[Nt]} \end{bmatrix} \mathbf{E}^{-1}$$

Solving linear dynamics problems

eigenvalues ϕ_1 and ϕ_2

$$\phi_1 = 1 - \frac{\omega^2}{4N^2} + \frac{\omega^2}{4N^2} \left(1 - \frac{16N^2}{\omega^2} \right)^{(1/2)}$$

$$\phi_2 = 1 - \frac{\omega^2}{4N^2} + \frac{\omega^2}{4N^2} \left(1 - \frac{16N^2}{\omega^2} \right)^{(1/2)}$$

$$\mathbf{E} = \begin{bmatrix} \frac{1}{4N} \left[1 - \left(1 - \frac{16N^2}{\omega^2} \right)^{1/2} \right] & \frac{1}{4N} \left[1 + \left(1 - \frac{16N^2}{\omega^2} \right)^{1/2} \right] \\ 1 & 1 \end{bmatrix}$$

Solving linear dynamics problems

Analytical solution of the PL system

$$X(t) = \frac{1}{a_{11} - a_{12}} [b_{11} \quad b_{12}] \mathbf{A} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix}$$

$$b_{11} = 1 - \frac{\omega^2}{2} \left(t - \frac{[Nt]}{N} \right)^2, \quad b_{12} = t - \frac{[Nt]}{N}$$

$$a_{11} = \frac{1}{4N} \left[1 - \left(1 - \frac{16N^2}{\omega^2} \right)^{1/2} \right], \quad a_{12} = \frac{1}{4N} \left[1 + \left(1 - \frac{16N^2}{\omega^2} \right)^{1/2} \right]$$

Solving linear dynamics problems

Useful limit results for $\lim_{N \rightarrow \infty} X(t)$

$$\lim_{N \rightarrow \infty} \phi_1^{[Nt]} = e^{it\omega}$$

$$\lim_{N \rightarrow \infty} \phi_2^{[Nt]} = e^{-it\omega}$$

$$\lim_{N \rightarrow \infty} \frac{a_{11}\phi_1^{[Nt]} - a_{12}\phi_2^{[Nt]}}{a_{11} - a_{12}} = \frac{e^{i\omega t} + e^{-i\omega t}}{2} = \cos(\omega t)$$

$$\lim_{N \rightarrow \infty} \frac{-a_{11}a_{12}\phi_1^{[Nt]} + a_{11}a_{12}\phi_2^{[Nt]}}{a_{11} - a_{12}} = \frac{1}{\omega} \sin(\omega t)$$

Solving linear dynamics problems

Therefore

$$\lim_{N \rightarrow \infty} X(t) = \begin{bmatrix} \cos(\omega t) & \frac{1}{\omega} \sin(\omega t) \\ 0 & 0 \end{bmatrix} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix}$$

i.e.

$$\lim_{N \rightarrow \infty} X(t) = x(t)$$

as expected

Solving linear dynamics problems

Continuous system

$$\ddot{x} + 2n\dot{x} + \omega^2 x = F \cos \Omega t$$

PL system

$$\ddot{X}_i + 2n\dot{X}_i + \omega^2 X_i = F \cos\left(\Omega \frac{i}{N}\right)$$

Solving linear dynamics problems

Solution

$$X(t) = e^{-n\left(t - \frac{[Nt]}{N}\right)} \begin{bmatrix} \cos\left(\xi t - \xi \frac{[Nt]}{N}\right) + \frac{n}{\xi} \sin\left(\xi t - \xi \frac{[Nt]}{N}\right) \\ \frac{1}{\xi} \sin\left(\xi t - \xi \frac{[Nt]}{N}\right) \end{bmatrix} \mathbf{w} \\ + \frac{F}{\omega^2} e^{-n\left(t - \frac{[Nt]}{N}\right)} \begin{bmatrix} e^{-n\left(t - \frac{[Nt]}{N}\right)} - \cos\left(\xi t - \xi \frac{[Nt]}{N}\right) \\ -\frac{n}{\xi} \sin\left(\xi t - \xi \frac{[Nt]}{N}\right) \end{bmatrix} \cos\left(\Omega \frac{[Nt]}{N}\right)$$

Solving linear dynamics problems

$$\mathbf{W} = e^{-n \frac{[Nt]}{N}} \mathbf{G}^{[Nt]} \begin{bmatrix} c_0 \\ d_0 \end{bmatrix} + \sum_{r=1}^{[Nt]} e^{-\frac{nr}{N}} \mathbf{G}^{r-1} \begin{bmatrix} e^{\frac{n}{N}} - \cos \frac{\xi}{N} - \frac{n}{\xi} \sin \frac{\xi}{N} \\ \left(\frac{n^2}{\xi} + \xi \right) \sin \frac{\xi}{N} \end{bmatrix} \frac{F}{\omega^2} \cos[\Omega([Nt] - r)]$$

$$\mathbf{G} = \begin{bmatrix} \cos \frac{\xi}{N} + \frac{n}{\xi} \sin \frac{\xi}{N} & \frac{1}{\xi} \sin \frac{\xi}{N} \\ -\left(\frac{n^2}{\xi} + \xi \right) \sin \frac{\xi}{N} & \cos \frac{\xi}{N} - \frac{n}{\xi} \sin \frac{\xi}{N} \end{bmatrix}$$

Solving linear dynamics problems

Therefore

$$\lim_{N \rightarrow \infty} X(t) = e^{-nt} \begin{bmatrix} \cos \xi t + \frac{n}{\xi} \sin \xi t & \frac{1}{\xi} \sin \xi t \end{bmatrix} \begin{bmatrix} d_0 \\ v_0 \end{bmatrix} \\ + \frac{F}{(\omega^2 - \Omega^2)^2 + (2n\Omega)^2} \begin{bmatrix} (\omega^2 - \Omega^2)(\cos \Omega t - e^{-nt} \cos \xi t) \\ + 2n\Omega \sin \Omega t - e^{-nt} (\omega^2 - \Omega^2) \frac{n}{\xi} \sin \xi t \end{bmatrix}$$

Solving nonlinear dynamics problems

- Numerical solutions
 - Taylor series
 - Runge-Kutta method

$$\begin{aligned} y_{n+1} &= y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4), & k_1 &= h f(t_n, y_n), \\ t_{n+1} &= t_n + h & k_2 &= h f\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \\ & & k_3 &= h f\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \\ & & k_4 &= h f(t_n + h, y_n + k_3). \end{aligned}$$

Characteristics of the Existing Numerical Approaches

- Discrete solutions
- Linearization and Taylor expansion → truncation with limited terms
- Damaged physical meaning and originality of the system considered

P-T Method for Oscillation Systems

General oscillation system:

$$\ddot{x} + 2c\dot{x} + \omega^2 x = \varphi(t) + \psi(t, x, \dot{x})$$

$\psi(t, x, \dot{x})$ linear or nonlinear function

Piecewise constant argument $[Nt]/N$:

$$\lim_{N \rightarrow \infty} \frac{[Nt]}{N} = t$$

Taylor series expansion

$$\ddot{x}_i + 2c\dot{x}_i + \omega^2 x_i = \varphi_i(t) + \psi_i \\ + \dot{\psi}_i t + \frac{1}{2!} \ddot{\psi}_i t^2 + \cdots + \frac{1}{n!} \psi_i^{(n)} t^n$$

in i th time interval, $[Nt]/N \leq t < ([Nt]+1)/N$.

- Continuous solution in the interval
- Combination of linear systems for $t > 0$

A damped linear oscillatory system,

$$\ddot{x}(t) + 2c\dot{x}(t) + a^2 x(t) = bx(t)$$

$$\ddot{x}_i(t) + 2c\dot{x}_i(t) + a^2 x_i(t) = bd_i + bv_i(t - \frac{[Nt]}{N})$$

Closed form solution:

$$x_i = e^{-c(t - \frac{[Nt]}{N})} \{B_1 \cos[\xi(t - \frac{[Nt]}{N})] + B_2 \sin[\xi(t - \frac{[Nt]}{N})]\} + A_1 + A_2(t - \frac{[Nt]}{N})$$

$$\xi^2 = a^2 - c^2$$

$$A_2 = \frac{1}{a^2}, \quad A_1 = \frac{1}{a^2} (bd_i - 2cA_2)$$

$$B_1 = d_1 - A_1, \quad B_2 = \frac{1}{\xi} (v_i + cB_1 - A_2)$$

Continuity conditions:

$$x_i\left(\frac{[Nt]}{N}\right) = x_{i-1}\left(\frac{[Nt]}{N}\right)$$

$$\dot{x}_i\left(\frac{[Nt]}{N}\right) = \dot{x}_{i-1}\left(\frac{[Nt]}{N}\right).$$

Recurrence relations:

$$d_{i+1} = e^{-\frac{c}{N}} \left(B_1 \cos \frac{\xi}{N} + B_2 \sin \frac{\xi}{N} \right) + A_1 + \frac{A_2}{N}$$

$$\begin{aligned} v_{i+1} = & -ce^{-\frac{c}{N}} \left(B_1 \cos \frac{\xi}{N} + B_2 \sin \frac{\xi}{N} \right) \\ & + e^{-\frac{c}{N}} \left(-\xi B_1 \sin \frac{\xi}{N} + \xi B_2 \cos \frac{\xi}{N} \right) + A_2 \end{aligned}$$

Comparison of the solutions of analytical, P-T and R-K methods:

Taylor expansion of P-T method Solution

$$\begin{aligned}x_i = & d_i + v_i \left(t - \frac{[Nt]}{N}\right) - \left[(a^2 - b) \frac{d_i}{2} + cv_i\right] \left(t - \frac{[Nt]}{N}\right)^2 \\& + \left[\frac{cd_i}{3} (a^2 - b) - \frac{v_i}{6} (a^2 - b - 4c^2)\right] \left(t - \frac{[Nt]}{N}\right)^3 \\& + \left[\frac{d_i}{24} (a^2 - b)(a^2 - 4c^2) \right. \\& \left. + \frac{cv_i}{6} \left(a^2 - \frac{b}{2} - 2c^2\right)\right] \left(t - \frac{[Nt]}{N}\right)^4 + \dots\end{aligned}$$

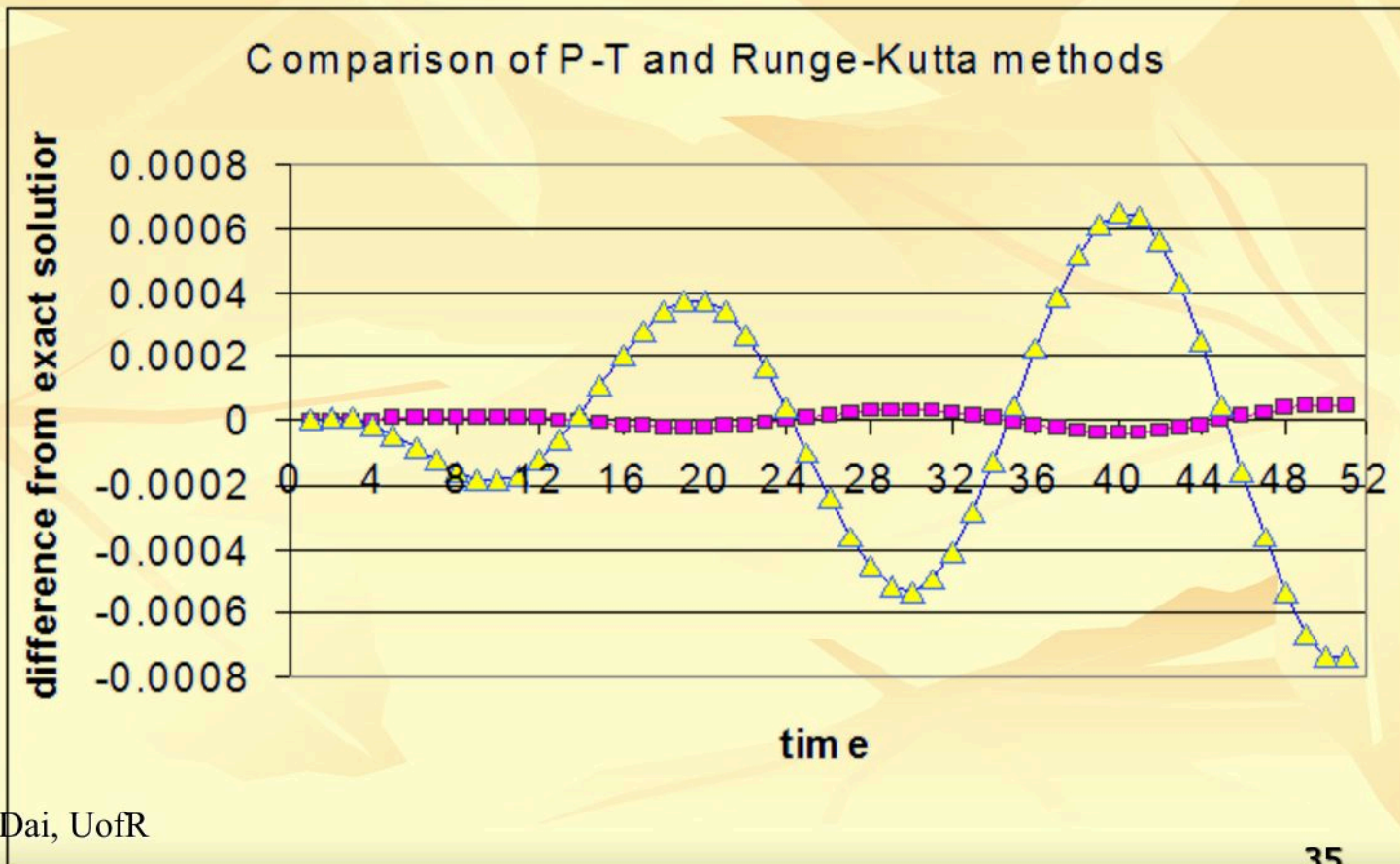
Exact solution:

$$\begin{aligned}x_i = & d_i + v_i \left(t - \frac{[Nt]}{N}\right) - [(a^2 - b) \frac{d_i}{2} + cv_i] \left(t - \frac{[Nt]}{N}\right)^2 \\& + \left[\frac{cd_i}{3} (a^2 - b) - \frac{v_i}{6} (a^2 - b - 4c^2)\right] \left(t - \frac{[Nt]}{N}\right)^3 \\& + \left[\frac{d_i}{24} (a^2 - b)(a^2 - b - 4c^2) \right. \\& \left. + \frac{cv_i}{6} (a^2 - b - 2c^2)\right] \left(t - \frac{[Nt]}{N}\right)^4 + \dots\end{aligned}$$

Difference is independent of the parameter N .

Solving nonlinear dynamics problems

P-T method vs. Runge-Kutta method



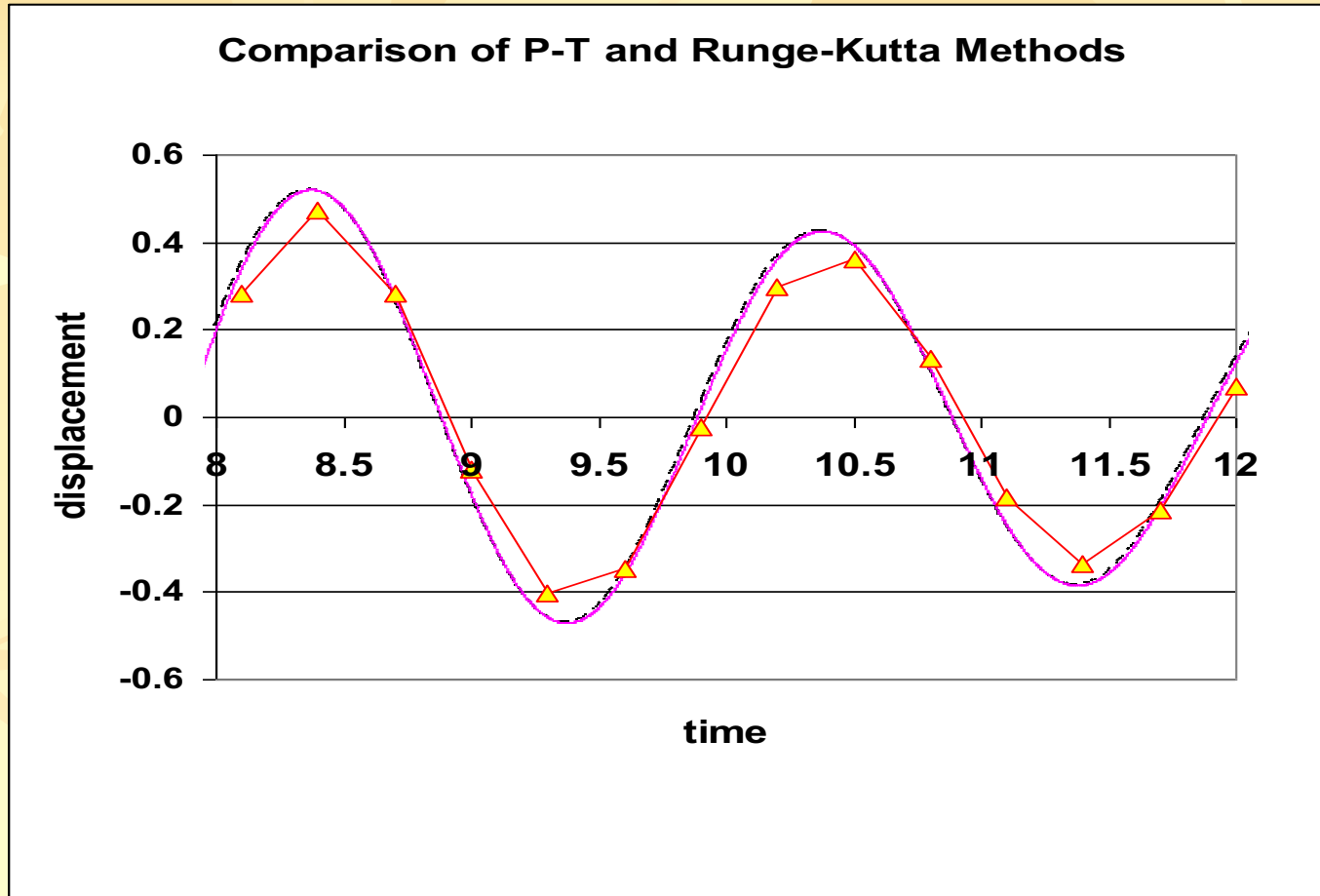
Solving nonlinear dynamics problems

- Numerical solutions
 - P-T method

$$\lim_{N \rightarrow \infty} \frac{[Nt]}{N} = t$$

$$\begin{aligned} \ddot{X}_i + 2c\dot{X}_i + \omega^2 X_i &= \varphi_i(t) + \psi_i \\ &+ \dot{\psi}_i t + \frac{1}{2!} \ddot{\psi}_i t^2 + \cdots + \frac{1}{n!} \psi_i^{(n)} t^n \end{aligned}$$

$$\ddot{x} + 0.2\dot{x} + 10x = x^3$$



Step length 0.3 P-T/ Runge-Kutta; 0.003 Runge-Kutta

Time	P-T Step 0.3	Runge-Kutta Step 0.3	Runge-Kutta Step 0.003
8.1	0.347724	0.281461	0.343268
8.4	0.517027	0.473953	0.518208
8.7	0.267069	0.28183	0.272421
9.0	-0.18254	-0.1204	-0.17695
9.3	-0.45882	-0.40096	-0.45817
9.6	-0.35455	-0.34717	-0.35839
9.9	0.027678	-0.02265	0.021947
10.2	0.364802	0.298642	0.362458
10.5	0.390918	0.362595	0.393205
10.8	0.1028	0.135726	0.107953
11.1	-0.25117	-0.18411	-0.24755
11.4	-0.3827	-0.33687	-0.38347
11.7	-0.20041	-0.2129	-0.20458
12	0.132724	0.072054	0.128444

Conclusion

- Nonlinear phenomena → more close to reality or nature
- Linearization leads to simplification and may miss responses such as chaos or quasiperiodicity
- Necessity for accurate and reliable analyses
- P-T method produces more accurate and reliable solutions in comparing with that of Runge-Kutta method, with about the same CPU time.

Conclusion

- Fully understand nonlinear systems and their responses eventually rely on analytical solutions
- Nonlinear theories → foundation for further experimental and numerical analyses
- Future of nonlinear dynamics relay on innovative theory, methods of analysis and solutions

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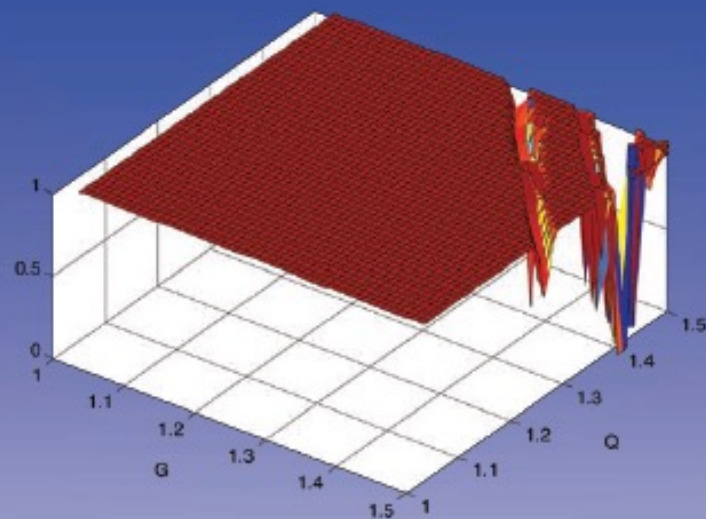
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Nonlinear Dynamics of Piecewise Constant Systems

Nonlinear Dynamics of Piecewise Constant Systems and Implementation of Piecewise Constant Arguments



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